The Islamia University of Bahawalpur
Department of Information Technology

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E-Commerce Application Development

Chapter 11

Shipping, Tax and Zones

Part III — From Basket to Order

Taught in Laboratory	Week 9
	Geo zones and GST; a voucher applied in the wrong order
Assessment	**Assignment 1 due**

This is a class extract. It is *Chapter 11* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

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Chapter 11

Shipping, Tax and Zones

Learning Outcomes

By the end of this chapter you will be able to:

- **Define** a geo zone, and **explain** why both shipping and tax are expressed in terms of one.
- **Distinguish** a tax rate, a tax class and a tax rule, and **state** what each one decides.
- **Compute** an order total by hand, including tax on shipping, and **reconcile** it against what the shop recorded.
- **Choose** a shipping method for a given shop, and **justify** the choice against what it costs to operate.
- **Apply** Pakistani sales tax to an online sale, and **identify** what you would have to verify before relying on it.

Before You Start

Chapter 10: the checkout sequence, and that an order copies its totals rather than recomputing them. A working checkout — the setting repaired in Chapter 10 must be in place or no order can be completed.

Key Terms in This Chapter

geo zone • zone • tax rate • tax class • tax rule • tax base • destination principle • flat rate • weight-based shipping • handling fee • order total line • sort order • GST • inclusive and exclusive pricing

11.1 Everything Here Depends on Where the Customer Is

A shop quotes one price for a product and then has to answer two questions that the price does not settle: *what does it cost to get this to you*, and *what tax is due on this sale*. Both answers depend on where "you" are, and neither can be stored on the product.

So the platform needs a way to talk about places. That is a **geo zone**.

Zone, Geo Zone

A **zone** is an administrative subdivision of a country — a province, a state, a county. Pakistan's zones are Punjab, Sindh, Khyber Pakhtunkhwa, Balochistan, and the territories. The platform ships a list of them for every country and you do not maintain it.

A **geo zone** is a *named set* of (country, zone) pairs that you do maintain, created because you want to treat those places alike. "Punjab and Islamabad", "everywhere except Gilgit-Baltistan", "the whole of Pakistan" are each a geo zone if a rule of yours applies to exactly that set.

The important property: a geo zone has no meaning of its own. It is a label you attach rules to. The same shop may have one geo zone for shipping and a different one for tax,

overlapping in any way, because the places where delivery costs the same are not the places where tax is the same.

</> The two geo zones the demonstration shop ships with

System › Localisation › Geo Zones shows them, and they are a good illustration precisely because they are different shapes.

UK Shipping	One row: country 222, zone 0. Zone 0 means *every* zone — the whole country in a single entry.
UK VAT Zone	Many rows: country 222 and one zone each. The same country, enumerated region by region.

Why the difference? Because delivery is priced for the country as a whole, while the tax zone was built to allow regions to be excluded one at a time. **Reading a geo zone's shape tells you what its author expected to change.**

11.2 Tax: Three Things, Not One

Most people arrive expecting a tax setting. There are three, they sit in a hierarchy, and conflating them is the commonest source of a wrong total.

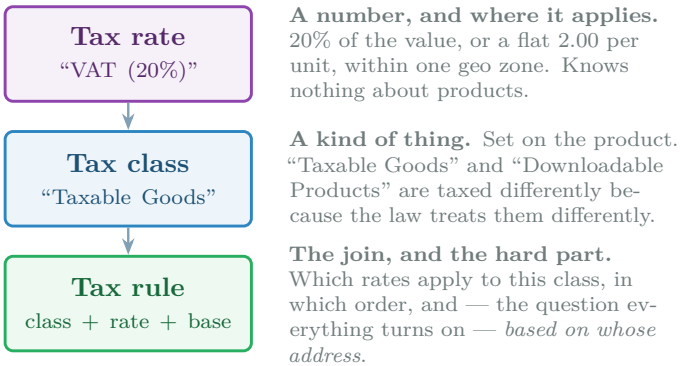


Figure 11.1: Tax in three layers. A rate is a number in a place; a class is a kind of product; a rule says which rates a class attracts, in what order, and which of the three addresses on the order decides.

The `based` column, which is the whole of cross-border tax

Every tax rule carries a **tax base**: `store`, `payment` or `shipping`. It names which address decides whether the rule applies.

- **`store`** — the shop's own address. Tax is the same for everyone. Simple, and correct only for a shop that sells domestically.
- **`payment`** — the billing address. Used where the law follows the buyer's financial residence.
- **`shipping`** — the delivery address. The **destination principle**: tax belongs where the goods are consumed. This is the usual answer for physical goods, and it is what the demonstration shop uses for its `Taxable Goods` rules.

One setting, three answers, and a shop that picks the wrong one is not slightly wrong — it is collecting the wrong tax from the wrong people and remitting it to the wrong authority. **System › Localisation › Taxes › Tax Rules** is where it is chosen, and it is the single most consequential dropdown in the admin panel.

11.3 An Order Total, Computed by Hand

The reliable way to understand a totals engine is to take one real order apart. This is the order placed by Chapter 10's harness: one MacBook at $500.00 and two iPhones at $101.00, delivered within the shop's own country.

Worked Example:

What the shop recorded. The rows of `oc_order_total`, in their own sort order:

Sub-Total	702.00	`sort_order` 1
Flat Shipping Rate	5.00	`sort_order` 3
Eco Tax (-2.00)	6.00	`sort_order` 5
VAT (20%)	141.40	`sort_order` 5
Total	**854.40**	`sort_order` 9

Step 1 — the sub-total. $500.00 + (2 \times 101.00) = 702.00$. Goods only; nothing else has been decided yet.

Step 2 — shipping. The flat rate is 5.00 regardless of what is in the basket. It is added as its own line, *before* tax, which matters for step 4.

Step 3 — the fixed-amount tax. "Eco Tax" is of type `F`: a flat 2.00 per unit, not a percentage. Three units were bought, so $3 \times 2.00 = 6.00$.

Step 4 — the percentage tax, and the part everyone gets wrong. 20% of *what*? Not of the sub-total:

$$0.20 \times 702.00 = 140.40 \neq 141.40$$

The difference is exactly 1.00, which is 20% of the 5.00 shipping charge:

$$0.20 \times (702.00 + 5.00) = 0.20 \times 707.00 = \mathbf{141.40}$$

Shipping is taxable, and the eco tax is not taxed again. Both of those are rule decisions, not arithmetic, and both would be different in another jurisdiction.

Step 5 — the total.

$$702.00 + 5.00 + 6.00 + 141.40 = \mathbf{854.40}$$

which is what the shop stored.

Two traps inside that one order

The `tax` column on an order line is *per unit*. `oc_order_product` recorded the iPhone line as quantity 2, total 202.00, tax **22.20**. That 22.20 is not the tax on the line; it is the tax on *one* iPhone — 20.20 of VAT plus 2.00 of eco tax. The line's tax is 44.40. A report that sums the `tax` column next to the `total` column adds a per-unit figure to a per-line figure and is wrong by a factor that changes with every basket.

The demonstration data is misleadingly named. "Eco Tax (-2.00)" reads as a two-unit *discount*. It is stored as rate +2.0000 and it **adds** 6.00 to this order. The name is the only thing negative about it. When you inherit a shop, the names in its tax table are the previous administrator's notes, not evidence.

11.4 Choosing How to Charge for Delivery

Five shipping methods ship with the platform. The choice is less about pricing than about what each one obliges you to maintain.

Method	Charges	What it obliges you to keep true
`flat`	One amount per order	Nothing. The cheapest to operate and the easiest to lose money on.
`item`	An amount per item	Nothing, but it punishes multi-item baskets, which is the opposite of what a shop wants.
`weight`	By total basket weight, in bands	**A weight on every product**, forever, including every product anyone adds later.
`free`	Nothing, above a threshold	A threshold chosen from your margin rather than from a competitor's.
`pickup`	Nothing; the customer collects	A real address and real opening hours.

How to choose, in one question

Ask: *what will this method be wrong about, and how often?*

A flat rate is wrong about every unusually heavy or distant order, and if those are 5% of your orders it is a rounding error you absorb. Weight-based shipping is right about every order and wrong whenever somebody adds a product and leaves the weight at zero — which is a defect that produces free delivery silently, and which Chapter 17 will not show you unless you ask it to.

The accurate method with a maintenance burden is worse than the approximate method with none, until the shop is big enough to carry the burden. That is the same argument Chapter 4 made about building versus extending, applied one level down.

11.5 Sales Tax on an Online Sale in Pakistan

Everything above is machinery. What a graduate working here will actually have to configure is Pakistani sales tax, and the machinery maps onto it directly.

Configuring a Pakistani shop

1. **Geo zone.** Create one named **Pakistan**, containing country Pakistan with zone 0 — every province. Sales tax on goods does not vary by province, so a single entry is correct and a list of provinces would be four more things to maintain.
2. **Tax rate.** One rate, type `P` (percentage), at the standard rate of **sales tax on goods** in force for the tax year. At the time of writing that standard rate is **18%**, but it has moved twice in recent years and moves with each Finance Act.
3. **Tax class.** **Taxable Goods**, applied to every physical product. Items on the Sixth Schedule are exempt and need a class of their own rather than a rate of zero, so that the exemption is visible in the data.
4. **Tax rule.** Base it on the **shipping** address. Goods are consumed where they are delivered.

5. **Display.** Decide whether displayed prices include tax. In Pakistan retail prices are customarily quoted inclusive, which means **System › Settings › Edit › Option** must be set to display prices with tax, or every product page will understate the price a customer pays.

What this section is not

It is a mapping from the platform's concepts to Pakistani tax concepts. It is **not** tax advice, and three things about it will go out of date:

- **The rate.** Verify the standard rate against the current Finance Act before configuring any shop. A handout printed in 2026 cannot be the authority for a rate set annually.
- **Services.** Sales tax on *services* is provincial — Sindh, Punjab, Khyber Pakhtunkhwa and Balochistan each levy their own at their own rate. A shop that sells only goods can ignore this; a shop that sells a service alongside goods cannot, and then the single geo zone above is no longer enough.
- **Marketplace obligations.** Where a shop sells through an online marketplace, the marketplace may carry withholding and reporting duties that a self-hosted shop does not. Chapter 22 returns to this with the E-Commerce Policy and FBR's registration requirements.

The professional habit this chapter is really teaching: configure the machinery so that a rate change is one field, and record where the number came from. A shop whose tax rate is hard-coded in three templates is a shop that will be wrong for a year after the next budget.

Common Pitfalls

- **Assuming tax applies to goods only.** It applied to shipping in the worked example, and 1.00 of the 141.40 was nobody's product.
- **Summing `oc_order_product.tax` as a line total.** It is per unit.
- **Trusting the names in an inherited tax table.** "Eco Tax (-2.00)" adds money.
- **Picking `store` as the tax base because it is first in the list.** It is correct only for a shop that never delivers anywhere else, and it is wrong silently.
- **Choosing weight-based shipping without an operational rule for entering weights.** The failure is free delivery, and it is invisible.
- **Changing a tax rate and expecting old orders to change.** They will not — an order copied its totals (Chapter 10), and that is the correct behaviour, not a bug.
- **Testing tax with the shop's own address.** Many rules only differ for a destination you have never tried.

Try It Yourself: Make your shop charge Pakistani tax, and prove the total

```
# --- 1. Look at what you have before changing it ---------------------
cd code
docker compose exec db mariadb -ushop -pshoppw opencart -e \
  "SELECT * FROM oc_geo_zone; SELECT * FROM oc_tax_rate;"

# --- 2. Place one order and take its total apart ---------------------
python m10_checkout.py --repeat 1
docker compose exec db mariadb -ushop -pshoppw opencart -e \
  "SELECT code,title,value FROM oc_order_total \
   WHERE order_id=(SELECT MAX(order_id) FROM oc_order) \
   ORDER BY sort_order;"
# Reproduce every line with a calculator before reading on.
```

```
# --- 3. Find the 1.00 ------------------------------------------------
# The VAT line is not 20% of the sub-total. Account for the difference
# exactly, and say which rule decision caused it.

# --- 4. Build a Pakistan geo zone ------------------------------------
# Admin: System > Localisation > Geo Zones > Add New.
# Pakistan, zone 0. One row.

# --- 5. Add the rate, the class and the rule -------------------------
# System > Localisation > Taxes > Tax Rates: percentage, current rate.
# System > Localisation > Taxes > Tax Classes: Taxable Goods.
# Base the rule on the SHIPPING address, and say why.

# --- 6. Disable the UK rules and order again -------------------------
python m10_checkout.py --repeat 1
# Predict the new total first. Then check the stored rows.

# --- 7. Break it deliberately ----------------------------------------
# Change the rule's base to "store" and order to the same address.
# Then order to a different country. Which one reveals the mistake?

# --- 8. Write the operating note -------------------------------------
# Five lines: where the rate came from, who changes it, when it is
# checked, and what else must change when it does.

# --- 9. Clean up -----------------------------------------------------
python m10_checkout.py --clean
```

What to notice. Step 7 is the lesson. A wrong tax base is invisible for every order delivered to the place you tested from, which is every order you will place while developing.

Checkpoint — Test Yourself

1. An order shows a sub-total of 702.00, shipping of 5.00 and a 20% VAT line of 141.40. Show where the 141.40 comes from, and state the two rule decisions the arithmetic reveals.
2. A shop in Lahore sells to customers across Pakistan and occasionally to the UAE. Its tax rule is based on `store`. Describe precisely what is wrong, which orders are affected, why nobody has noticed, and what you would change.
3. A colleague proposes weight-based shipping for a shop with 4,000 products, 600 of which have a weight of zero. Give the failure mode in terms of money, say why no error appears anywhere, and give one change that would make the failure visible.

Chapter Summary

- A geo zone is a named set of (country, zone) pairs. It has no meaning of its own; it is a label that shipping methods and tax rules attach to, and a shop's shipping zones need not match its tax zones.
- Tax is three things: a *rate* is a number in a place, a *class* is a kind of product, and a *rule* joins them and chooses which address decides.
- The `based` column — store, payment or shipping — is the single most consequential setting in the tax machinery. For physical goods the destination principle makes `shipping` the usual answer.
- Worked: 702.00 of goods + 5.00 shipping + 6.00 fixed eco tax + 141.40 VAT = 854.40, where the VAT is 20% of goods *plus shipping*, and the 1.00 difference is the whole lesson.

- `oc_order_product.tax` is per unit, not per line. Summing it beside `total` is wrong by the quantity.
- Shipping methods differ less in what they charge than in what they oblige you to keep true. Weight-based shipping requires a weight on every product forever, and fails as silent free delivery.
- For a Pakistani shop: one geo zone with zone 0, one percentage rate verified against the current Finance Act, a Taxable Goods class, and a rule based on the shipping address.
- Changing a rate changes future orders only. Orders copy their totals, and that is correct.

Review Questions

1. **(MCQ)** A tax rule's `based` column set to `shipping` means tax is decided by: (a) the shop's address; (b) the billing address; (c) the delivery address; (d) the shipping method chosen.

2. **(MCQ)** In the worked order, the VAT line exceeded 20% of the sub-total by exactly 1.00 because: (a) rounding; (b) shipping is taxable; (c) the eco tax was taxed; (d) there were three units.

3. **(Short)** Explain the difference between a tax rate and a tax class, with one example of two products that share a rate but not a class.

4. **(Short)** Why does a shop ever need more than one geo zone? Give a concrete case where the shipping zone and the tax zone must differ.

5. **(Short)** Explain why exempt products should be given their own tax class rather than a tax rate of zero.

6. **(Applied)** A shop quotes prices inclusive of tax and then changes the tax rate from 17% to 18%. Say exactly what must change in the configuration, what happens to existing orders, what happens to displayed prices, and what the shop must decide about who absorbs the 1%.

7. **(Applied)** Design the shipping configuration for a Bahawalpur shop selling 200 products weighing between 50 g and 12 kg, delivering nationwide, with free delivery intended above Rs 5,000. State your methods, your zones, your thresholds, what each one obliges somebody to maintain, and how you would detect the day it starts losing money.